

Research Blueprint: Multi-Physics PINN for 12-Month PV Predictive Maintenance

1. Executive Summary

This project bridges advanced physics-informed machine learning with real-world solar energy operations. You are building a **Physics-Informed Digital Twin (DT)** that uses a **Physics-Informed Neural Network (PINN)** to predict photovoltaic (PV) power degradation **12 months into the future** under varying climate scenarios, while diagnosing internal physical fault mechanisms before failure occurs.

2. The Core Problem & Research Gap

The Problem in Industry

- Solar panels degrade due to heat, UV exposure, micro-cracks, and solder fatigue.
- Current monitoring systems only measure surface parameters (Voltage, Current, Temperature, Irradiance). They cannot "see" internal physical degradation (R_s - series resistance, R_{sh} - shunt resistance) until a catastrophic failure or significant power loss happens.

The Research Gap in Existing Literature

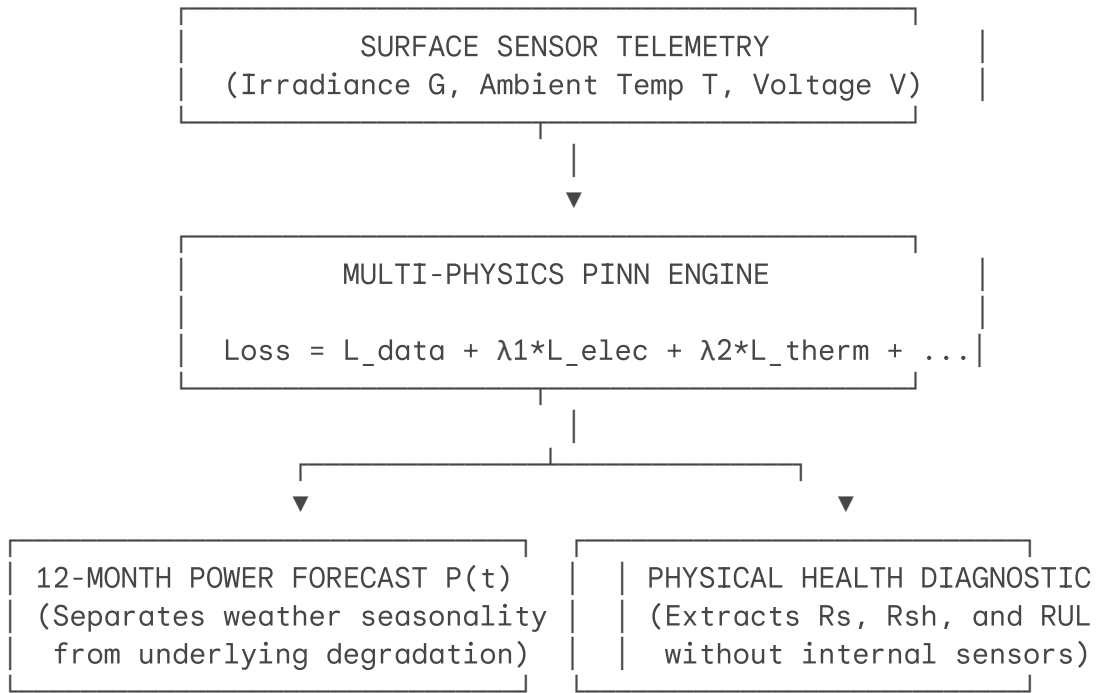
- Black-Box ML Models (LSTMs, XGBoost, Transformers):**
 - Require huge amounts of training data.
 - Fail when forecasting long horizons (12 months) under unseen climate conditions (e.g., a +2°C heatwave or heavy cloud cover year) because they lack physical constraints and drift into impossible predictions (e.g., negative efficiency).
- Traditional Physical Solvers:**
 - Require exact internal parameter measurements that sensors cannot provide in real time.
 - Too computationally slow for real-time digital twin monitoring.
- Standard PINN Research:**
 - Most existing papers use simple 1-step or short-term static equations. Very few couple **electrical physics + thermal physics + physical aging ODEs** over a 12-month time horizon.

3. Your Research Novelty (The "Paper" Angle)

Your contribution is a **Multi-Physics Dynamic Parameter Estimation PINN**.

Instead of treating internal cell parameters as static constants, your PINN solves an **inverse physics problem**: it extracts time-varying internal physical parameters ($R_s(t)$ and $R_{sh}(t)$)

directly from noisy, basic surface telemetry.



The Embedded Physical Constraints ($L_{physics}$)

Your neural network's loss function will enforce three coupled physical laws:

1. Electrical Physics (Single-Diode Equation):

$$\mathcal{L}_{elec} = I(t) - \left(I_{ph}(G) - I_0 \left[\exp \left(\frac{V(t) + I(t)R_s(t)}{nV_t} \right) - 1 \right] - \frac{V(t) + I(t)R_s(t)}{R_{sh}(t)} \right),$$

2. Thermal Dynamics (Energy Balance):

$$\mathcal{L}_{thermal} = C_{th} \frac{dT}{dt} - (Q_{in}(G) - P_{elec}(V, I) - h_{loss}(T_{cell} - T_{ambient}))$$

3. Physical Aging Law (Arrhenius Thermal Degradation ODE):

$$\mathcal{L}_{aging} = \frac{dR_s(t)}{dt} - A \cdot \exp \left(-\frac{E_a}{k \cdot T_{cell}(t)} \right)$$

Why this novelty is strong: You are proving that by embedding Arrhenius thermal degradation laws and diode physics into a PINN, you can accurately forecast 12-month degradation trajectories even when sensor data is sparse or noisy.

4. The 12-Month Future Prediction Angle (The "Climate" Aspect)

To predict 12 months ahead, your model separates **two distinct frequency signals**:

- **High-Frequency Seasonal Wave:** Short-term weather/climate variations (summer peak vs. winter drop, cloud cover, daily temperature cycles).
- **Low-Frequency Degradation Trend:** The irreversible physical degradation curve driven by thermal stress ($R_s(t)$ growth).

Climate Scenario Evaluation

You will evaluate your PINN under three distinct 12-month climate profiles:

1. **Baseline Meteorological Year:** Historical 10-year average solar irradiance and ambient temperature profiles.
2. **Climate Shift / Heatwave Scenario (+1.5°C to +2.0°C):** Demonstrates how higher thermal stress accelerates physical R_s growth in the Arrhenius ODE.
3. **Soiling / Low-Irradiance Scenario (-10% Irradiance):** Demonstrates model resilience under reduced sunlight conditions.

5. The Deliverable (The "Supervisor Dashboard" Angle)

To make your research tangible for your supervisor and final submission, you package the model into a **PV Digital Twin Predictive Maintenance Dashboard**.

Dashboard Key Modules:

1. **Interactive Controls:**
 - **Climate Scenario Dropdown:** Standard Year vs. Heatwave (+2°C) vs. High Soiling.
 - **Noise Level Slider (0% to 20%):** Demonstrates model robustness.
 - **Fault Injection Selector:** Inject Hotspot, Solder Fatigue, or Shunt Degradation.
2. **12-Month Power Forecast & Benchmark Chart:**
 - Plots **Actual/Target Data**, **PINN 12-Month Projection**, and **Standard LSTM Baseline**.
 - Shows how the standard LSTM drifts wildly under noisy/unseen weather, while the PINN stays physically bounded.
3. **Internal Physics Health Gauges:**
 - Displays dynamically extracted R_s (Series Resistance in Ohms) and R_{sh} (Shunt Resistance in Ohms).
 - **Remaining Useful Life (RUL)** metric in years/months.
4. **Actionable Maintenance Alert Panel:**
 - Automatically calculates the exact future month when R_s will cross the safe operating threshold ($R_s > 0.15 \Omega$).
 - *Example Alert:* "Month 8 Hotspot Warning: Series Resistance projected to exceed safe limit. Schedule thermal camera inspection in Month 7."

6. Implementation Checklist & Timeline

Phase	Core Goal	Key Deliverables
Phase 1: Lit Review	Review 10–15 core PINN/PV papers	Literature Summary Matrix identifying the gap
Phase 2: Baseline	Run existing repository code	Verified benchmark environment
Phase 3: Model Dev	Implement PyTorch PINN with Arrhenius $R_s(t)$ ODE	Loss function with coupled $L_{data} + L_{elec} + L_{therm} + L_{aging}$
Phase 4: Experiments	Benchmark against LSTM under noise & 12-mo climate scenarios	RMSE, MAE, R^2 performance comparison tables
Phase 5: Dashboard	Build Streamlit / Plotly UI	Interactive Digital Twin Dashboard
Phase 6: Paper	Draft research manuscript	Abstract, Methodology, Results, Discussion, Conclusion